Year 13 Further Maths - Dimensional analysis

Similar questions - Finding formulae and checking consistency

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 The frequency f of a vibrating string is thought to depend on its tension T, its length l and its mass per unit length μ, so that $f = kT^{\alpha}l^{\beta}\mu^{\gamma}$ where k is a dimensionless constant.

(a) Write down the dimensions of f, T and μ. **[3]**

(b) Use dimensional analysis to find α, β and γ. **[5]**

(c) Describe the effect on f of quadrupling the tension while keeping l and μ constant. **[1]**

2 For each of the following suggested formulae, determine whether it is dimensionally consistent. Here T is time, l length, g acceleration due to gravity, ρ density, A area, v speed and P power.

(a) $T = 2\pi\sqrt{\dfrac{g}{l}}$ **[3]**

(b) $P = \frac{1}{2}\rho A v^3$ **[4]**

3 The speed v of waves on the surface of deep or shallow water is modelled as depending on the acceleration due to gravity g, the depth h and the density of water ρ: $v = kg^{\alpha}h^{\beta}\rho^{\gamma}$ with k dimensionless.

(a) Use dimensional analysis to find α, β and γ. **[5]**

(b) Given that $k = 1$, find the speed of a wave in an ocean of depth 4000 m, and the time in hours for it to cross 5000 km. **[3]**

(c) Explain why the model cannot be used to find the value of k. **[1]**

4 The volume flow rate Q of a viscous fluid through a narrow pipe of radius r is thought to depend on r, the viscosity η of the fluid (dimensions $\mathrm{ML^{-1}T^{-1}}$), and the pressure gradient G along the pipe (force per unit area per unit length).

(a) Find the dimensions of Q and of G. **[3]**

(b) Assuming $Q = kr^{\alpha}\eta^{\beta}G^{\gamma}$, find α, β and γ. **[5]**

(c) The radius of the pipe is doubled and nothing else changes. By what factor does the flow rate change? **[1]**

Question 1

(a) $[f] = \mathrm{T}^{-1}$, $[T] = \mathrm{MLT}^{-2}$, $[\mu] = \mathrm{ML}^{-1}$

(b) $\mathrm{T}^{-1} = \mathrm{M}^{\alpha+\gamma}\mathrm{L}^{\alpha+\beta-\gamma}\mathrm{T}^{-2\alpha}$. T: $\alpha = \frac{1}{2}$; M: $\gamma = -\frac{1}{2}$; L: $\frac{1}{2} + \beta + \frac{1}{2} = 0$, $\beta = -1$. $f = \dfrac{k}{l}\sqrt{\dfrac{T}{\mu}}$

(c) f doubles

Question 2

(a) $\left[\sqrt{g/l}\right] = \left(\mathrm{LT}^{-2}/\mathrm{L}\right)^{1/2} = \mathrm{T}^{-1} \neq \mathrm{T}$, so it is not consistent (it should be $\sqrt{l/g}$)

(b) $[\rho A v^3] = \mathrm{ML}^{-3} \cdot \mathrm{L}^2 \cdot \mathrm{L}^3\mathrm{T}^{-3} = \mathrm{ML}^2\mathrm{T}^{-3}$, which equals $[P] = [\text{force} \times \text{speed}] = \mathrm{MLT}^{-2} \cdot \mathrm{LT}^{-1}$. Consistent

Question 3

(a) $\mathrm{LT}^{-1} = (\mathrm{LT}^{-2})^{\alpha}\mathrm{L}^{\beta}(\mathrm{ML}^{-3})^{\gamma}$. M: $\gamma = 0$; T: $\alpha = \frac{1}{2}$; L: $\frac{1}{2} + \beta = 1$, $\beta = \frac{1}{2}$. $v = k\sqrt{gh}$

(b) $v = \sqrt{9.8 \times 4000} = 198\ \mathrm{m\,s^{-1}}$; time $= \frac{5\times10^6}{198} = 25\,300\ \mathrm{s} = 7.01\ \mathrm{h}$

(c) Dimensional analysis only fixes the powers; a dimensionless constant cannot be found from dimensions, only by experiment

Question 4

(a) $[Q] = \mathrm{L}^3\mathrm{T}^{-1}$; $[G] = \mathrm{ML}^{-1}\mathrm{T}^{-2} \cdot \mathrm{L}^{-1} = \mathrm{ML}^{-2}\mathrm{T}^{-2}$

(b) M: $\beta + \gamma = 0$; T: $-\beta - 2\gamma = -1$; so $\gamma = 1$, $\beta = -1$; L: $\alpha - \beta - 2\gamma = 3$, $\alpha = 4$. $Q = \dfrac{kr^4G}{\eta}$

(c) Factor of $2^4 = 16$